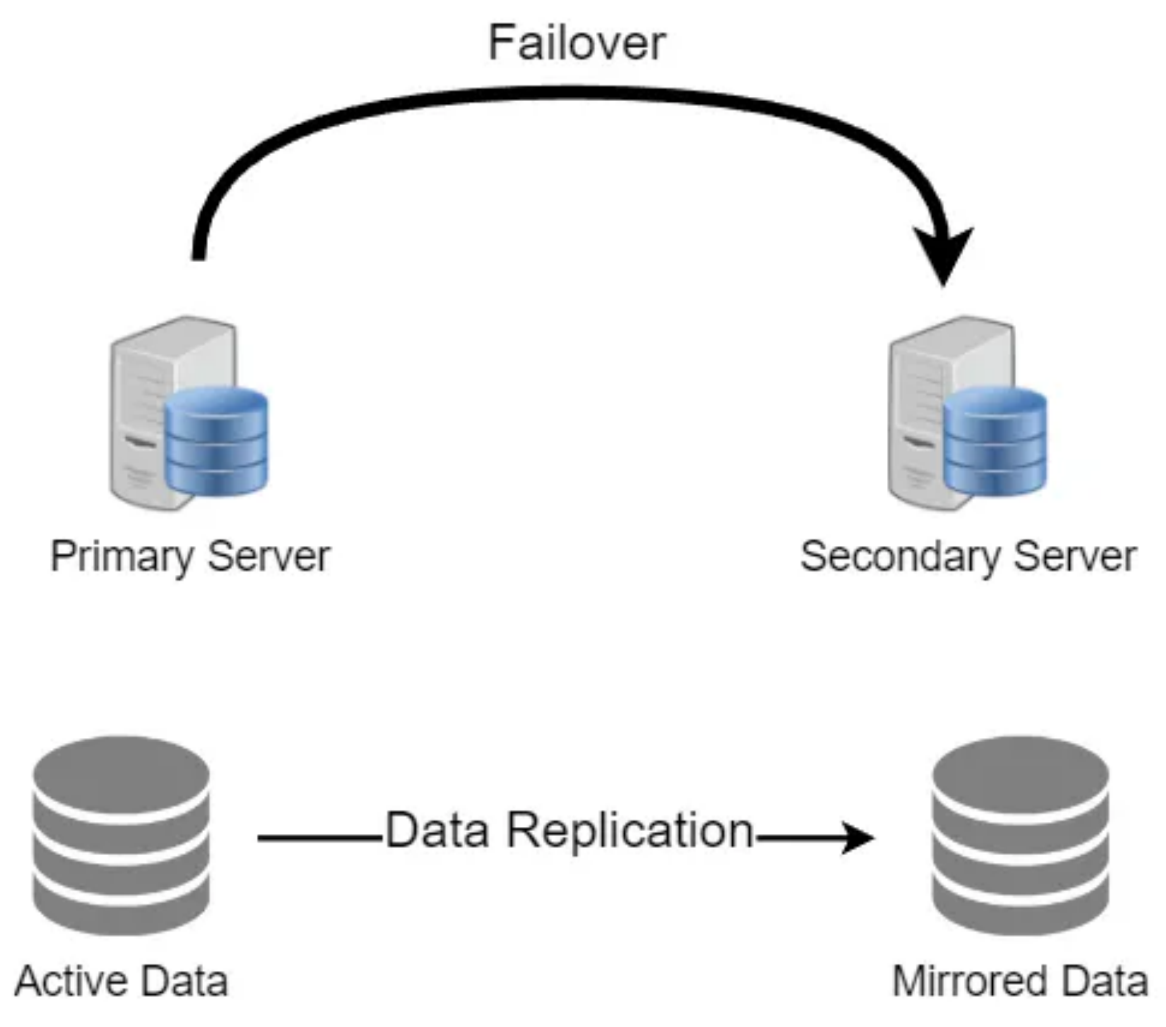
Redundancy and Replication

Redundancy is the duplication of critical components or functions of a system with the intention of increasing the reliability of the system, usually in the form of a backup or fail-safe, or to improve actual system performance. For example, if there is only one copy of a file stored on a single server, then losing that server means losing the file. Since losing data is seldom a good thing, we can create duplicate or redundant copies of the file to solve this problem.

Redundancy plays a key role in removing the single points of failure in the system and provides backups if needed in a crisis. For example, if we have two instances of a service running in production and one fails, the system can failover to the other one.



Database **replication** is the process of copying and synchronizing data from one database to one or more additional databases. This is commonly used in distributed systems where multiple copies of the same data are required to ensure data availability, fault tolerance, and scalability.

Replication is widely used in many database management systems (DBMS), usually with a primary-replica relationship between the original and the copies. The primary server gets all the updates, which then ripple through to the replica servers. Each replica outputs a message stating that it has received the update successfully, thus allowing the sending of subsequent updates.

Here are the top three typical database replication strategies:

Synchronous replication is a type of database replication where changes made to the primary database are immediately replicated to the replica databases before the write operation is considered complete. In other words, the primary database waits for the replica databases to confirm that they have received and processed the changes before the write operation is acknowledged.

In synchronous replication, there is a strong consistency between the primary and replica databases, as all changes made to the primary database are immediately reflected in the replica databases. This ensures that the data is consistent across all databases and reduces the risk of data loss or inconsistency.

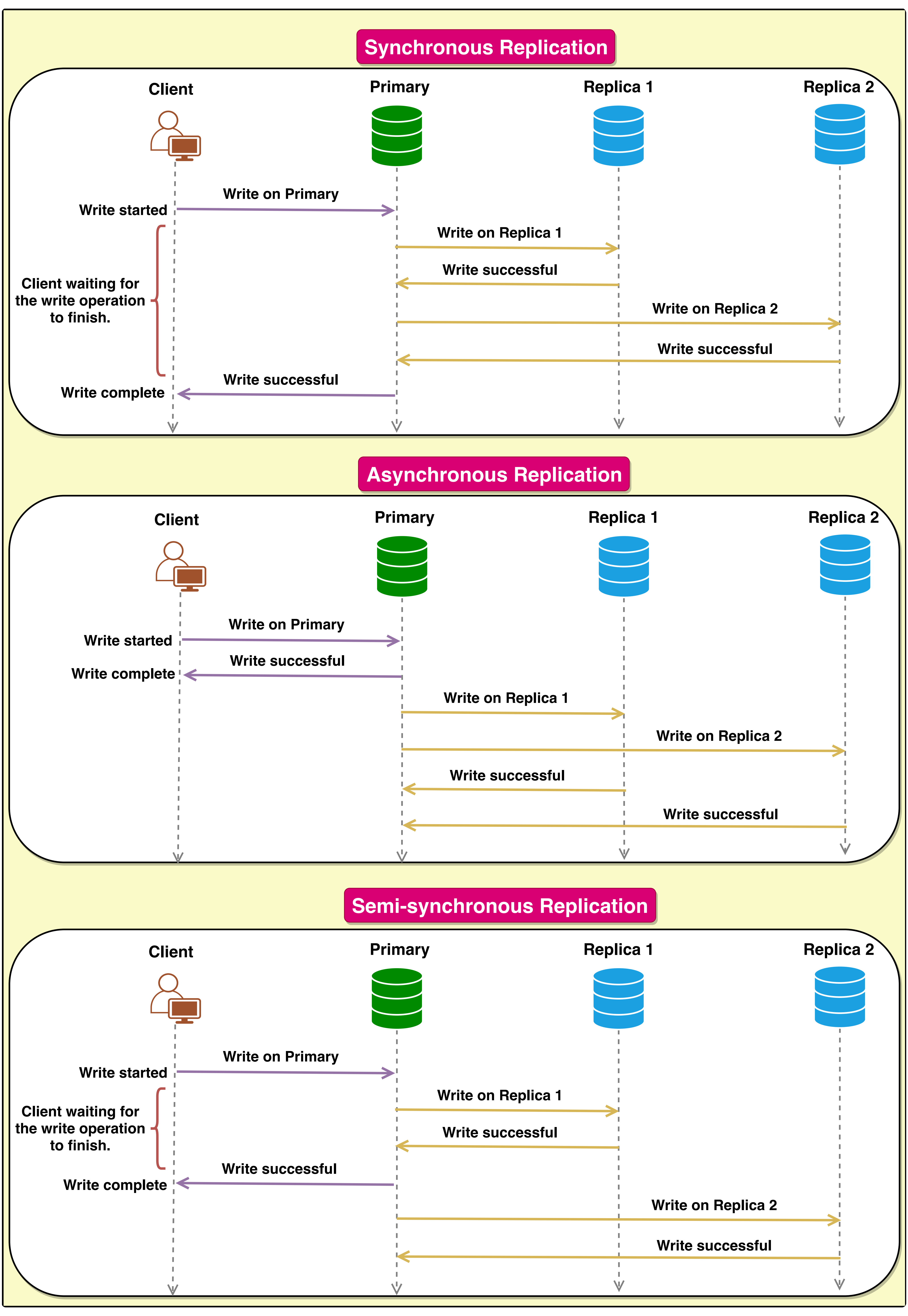
Asynchronous replication is a type of database replication where changes made to the primary database are not immediately replicated to the replica databases. Instead, the changes are queued and replicated to the replicas at a later time.

In asynchronous replication, there is a delay between the write operation on the primary database and the update on the replica databases. This delay can result in temporary inconsistencies between the primary and replica databases, as the data on the replica databases may not immediately reflect the changes made to the primary database.

However, asynchronous replication can also have performance benefits, as write operations can be completed quickly without waiting for confirmation from the replica databases. In addition, if one or more replica databases are unavailable, the write operation can still be completed on the primary database, ensuring that the system remains available.

Semi-synchronous replication is a type of database replication that combines elements of both synchronous and asynchronous replication. In semi-synchronous replication, changes made to the primary database are immediately replicated to at least one replica database, while other replicas may be updated asynchronously.

In semi-synchronous replication, the write operation on the primary is not considered complete until at least one replica database has confirmed that it has received and processed the changes. This ensures that there is some level of strong consistency between the primary and replica databases, while also providing improved performance compared to fully synchronous replication.



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